A Uniform Response Algorithm and Alternating-Segment Compression for Four-Column Gravity Connect Four

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Abstract

We organize covering strategies for four-column gravity Connect Four into one deterministic response rule for both players and every finite height. The rule reads the complete current board, the controlled color, and the opponent's last column. It stores no phase, reserved target, or active-switch counter, and does not search a future game tree. Its proof separates geometric position certificates, covering closure, full-column exits, and the initial uncertified states. The precise selection order matters. A shared executable Lean 4 function has been proved non-losing in both roles, with an optimal active-switch budget and an explicit bridge to legal-history prefix semantics. A further paper proof bounds the total number of maximal alternating segments in every legal candidate board by nine. A three-stream interval intersection replaces segment-combination enumeration for the winning-target subproblem with a linear scan. The complete compressed engine, its refinement to the accepted response function, and its cost theorem have not been formalized; finite comparison tests do not establish these claims.

Keywords: strategy certificate; position invariant; alternating segment; interval intersection; Lean 4.

Relationship to the mathematical paper

The companion mathematical paper [4] proves empty-board draws at widths one to four and exact active-switch budgets. The present paper concerns uniform execution on finite four-column boards and compressed representations. Pairing, Claimeven, and even-capacity release have precedents in Allis [1]. The present implementation chooses a move through current-position checks rather than future-tree search or a height-indexed strategy table. Infinite and semi-infinite games have their own literature [2]. This finite algorithm does not accept an infinite capacity or assign a parity to it. The executable definition and proofs are supplied in the repository [5], together with the interval-scan implementation and finite checks [6].

1 Game, cost, and algorithm input

1.1 Legal play and first-win stopping

The board has four columns and a finite height $H \geq 0$. Rows start at one at the floor. A move drops a stone into the lowest vacant square of a nonfull column. Horizontal, vertical, and either diagonal fours win. Play stops at the first win or a full board; the losing player has no corrective reply. Height zero is an immediate draw.

We use F, S for the first and second players and write column words bottom to top. For local certificates, P is the responding player and Q the opponent. Out-of-board lines and targets are excluded. A fixed non-losing policy must handle every legal opponent continuation, not merely a self-play run.

1.2 Switch cost

A reply in a different column is active when the opponent's last column remains nonfull, and passive when the opponent just filled it. Covers have zero cost; the first player's opening is free. A reply that wins is still charged before termination. A non-losing policy's budget is its worst-case active count over all opponent continuations, minimized over all such policies when discussing optimality.

1.3 Inputs and lack of phase memory

For $H > 0$ the first player opens at $(1, 1)$. Thereafter the same response function serves both roles. It reads the board after the opponent's move, the controlled color, and the previous move's column c. Its row r is the current height of c.

All targets, exceptions, and certificates are recomputed from the current board. No extra phase label, old target, old certificate, or used-budget counter is stored. "Local" here means a current-board and one-candidate-move check, not a fixed-radius neighborhood. Certificates below apply after a hypothetical or actual P reply, with Q to move.

2 Support exceptions and common safety conditions

2.1 Support descent

Lemma 1. If a nonbottom nonvertical Q four has a P stone immediately below every stone, it cannot be a first Q win.

Proof. Translating the line down one row gives a P four. All its stones existed before the last Q move, by gravity. ▫

2.2 Complete exceptions and unresolved lines

In a nonterminal candidate position, let E contain every nonbottom Q stone directly above Q, and the next empty square of every nonfull Q-topped column. The second type is a potential support exception. Do not add a square above a full column. Require that each nonfull top run of Q has length at most two.

Let U be the valid nonvertical four-cell lines meeting the bottom row or E and containing no P stone. Other nonvertical lines are already blocked or excluded by support descent. At heights at most three the two bottom diagonals do not exist. An already won or full board is handled as terminal, not passed off as a nonterminal certificate.

2.3 What covering preserves

A Q move followed by a cover creates no exception outside the old E: a potential exception may become actual and must remain recorded. A Q move filling a column, followed by a P stone elsewhere, also creates no new exception outside E. A top Q run grows from at most two to at most three before being capped by P or the ceiling. Existing blockers persist. Consequently the newly relevant unresolved lines form a subset of the old ones.

What still needs proof is that every old unresolved line remains protected against the opponent's move itself, before the reply. The next certificates provide this protection.

3 Three decidable certificate classes

Write h_j for the current height of column j. Every certificate includes the common nonterminal, boundedness, and top-run conditions of Section 2.

3.1 Class A: permanent safety or one target

Accept class A if one of the following holds:

- **A0:** U is empty.
- **A1:** There is an in-board empty target (j, g) with positive even gap $g - h_j$, even total capacity outside column j, and every line of U passes through (j, g).
- **A2:** The target has the same capacity and gap conditions, and three fixed current P stones would form a four with a P stone at the target. Every line of U passes through some square in column j at or above row g.

A2 is a geometric test, not permission to drop a stone over empty squares. Its other three stones must already exist. All unresolved lines must be protected by the same target.

Lemma 2 (A safety and closure). From an A-certified position the opponent cannot win first on the next move. If another reply is needed, a legal cover yields another A certificate or a safe terminal result. If the opponent fills its column, some legal reply has the same property.

Proof. For A0, bottom and exception lines are permanently blocked; support descent and the top-run condition handle the rest. A cover or any legal post-fill move preserves these facts.

For A1 and A2, a positive even target gap prevents Q from reaching the target: a visit reaches at most the square below it, with room for a cover. The target and the higher squares in its column remain empty until P acquires it. Thus the protected lines cannot complete first. Covers preserve the gap parity or acquire the target.

An opponent fill cannot occur in the still-pending target column. It occurs outside, whose positive even capacity before the move was at least two. There is therefore a legal outside reply. Such a reply preserves the target and all fixed stones. Before acquisition retain the certificate; on A1 acquisition all unresolved lines become blocked, giving A0; on A2 acquisition P wins. Any earlier P win or a full-board draw also ends the process safely. ▫

The implementation need not remember the old target. The old certificate ensures safety of the next opponent move and existence of a good successor; the selected successor's certificate takes over.

3.2 Class B: zero or two odd-capacity columns

Let

$$R = \{j : H - h_j \text{ is odd}\}.$$

Let G contain all current P squares and every empty in-board square (j, g) at a positive even gap from h_j. Such an empty square remains empty or becomes P under repeated covering.

A B certificate requires:

1. $|R|$ is zero or two.
2. Every line in U meets G.
3. If $R = \{f, t\}$, check **both ordered choices** (f, t) and (t, f): every line of U meets the set consisting of G outside the prospective receiving column t, together with (t, H).

The receiving column's top square is currently empty because its capacity is positive and odd. Condition 3 is not optional: its old even-gap guards may cease to be guards after an unpaired transfer.

Lemma 3 (B safety and closure). A B certificate excludes a first opponent win. Covering preserves B or reaches a safe terminal state. If the opponent fills a column, there were two odd-capacity columns, and replying in the other one is legal and yields a B certificate with $R = \varnothing$, or a safe terminal state.

Proof. With no odd capacities, every playable column has at least two vacancies at a Q turn, so covering is always possible. The guards remain empty or become P. With two odd capacities, complete covering rounds preserve R and the guard conditions; unresolved lines can only disappear.

The first column filled by Q is some $f \in R$, and that move is still protected by the old G. The other column t has positive odd capacity. A P move there makes every remaining capacity even. Old guards outside t persist; old guards in the now-full f have become fixed P stones. Guards in t may be lost, but (t, H) becomes either a P stone or an empty square at positive even gap. Condition 3 supplies precisely the surviving protection for every unresolved line. No new exception appears, so B holds with no odd capacities. ▫

No remembered choice of R or "already transferred" flag is required.

3.3 Class C: empty middle columns

C requires all four capacities to be odd, both middle columns empty, and both exterior columns to be alternating even-length words $(QP)^a$, $(QP)^d$, with $a, d \geq 0$ and $a + d > 0$. These conditions imply odd $H \geq 3$.

Lemma 4 (C safety and exits). Covering a non-filling outside move preserves C or safely terminates. When the opponent first enters a middle column or fills an outside column, there is a legal B-certified or terminal reply. The event move itself is safe.

Proof. Before the event both middle columns are empty, and after it at least one remains empty. No nonvertical four is possible; exterior words alternate and a newly entered middle column contains one stone, excluding vertical fours.

On entry into column two, reply at the bottom of column three, and conversely. The middle heights become one and the odd capacities are the exterior pair. The bottom horizontal contains P. The sole potential exception is the second square above the opponent's middle bottom stone; its horizontal is blocked by the second square of a nonempty exterior word. The falling and rising bottom diagonals are guarded by $(2, 3)$ and $(3, 3)$. These guards lie outside either possible receiving exterior column, satisfying both B exits. At $H = 3$ the diagonals are absent.

When column one fills, reply at $(3, 1)$. For $H \geq 5$ the fixed $P(1, 4)$ blocks the falling bottom diagonal and $(3, 3)$ guards the rising one. The odd capacities are columns two and four, neither containing that guard. A full Q-topped column adds no out-of-board exception. Reflect for a fill in column four, replying at $(2, 1)$. At $H = 3$ only the bottom horizontal remains relevant and is blocked. All replies enter an empty middle column and are legal. ▫

This event table proves existence of qualified replies; the algorithm does not store it as a phase table.

3.4 Role parity

After a first-player reply the total number of stones, and hence vacancies, is odd. Thus the number of odd-capacity columns is odd, excluding both B and C. A first-player certificate must be A. After a second-player reply the total vacancy count is even and all three classes can be checked.

The even outside-capacity condition of A1/A2 implies odd target-column capacity for the first player and even target-column capacity for the second. The implementation checks the same condition in both roles instead of branching on this consequence.

4 One deterministic selection order

4.1 Qualified replies

After the opponent's move, first stop if it wins or fills the board. Otherwise, hypothetically drop P in each legal column j. Call j qualified if the candidate wins for P, is a no-win full board, or is nonterminal and has an A, B, or C certificate. A hypothetical move is not an actual move, and never repairs an existing opponent win.

4.2 Priority rules

Let c, r be the opponent's last column and row.

Step 1. If c is still playable and either its cover is qualified or both bottom corners $(1, 1), (4, 1)$ already belong to P, cover in c. Playability is required for both alternatives.

Step 2. Otherwise, if qualified replies exist, choose an immediately winning one if possible, breaking ties to the left. If none wins immediately, prefer qualified columns whose capacity before the reply is odd, then choose the leftmost.

Step 3. If neither earlier step selected a move, then on reachable histories c is still playable. Cover if $r \geq 3$. If $r \leq 2$, score each legal candidate by the number of bottom lines already containing P after it; maximize this score, preferring a cover on ties and then the leftmost column.

To totalize the function on other nonterminal legal inputs, Step 3 first checks whether c is full and then returns the leftmost legal column if so. The proof shows this fallback is unreachable from the specified openings. There is no test of history compatibility in the executable rule.

The only separate initialization is the first player's move at $(1, 1)$ for $H > 0$. The remaining response code is identical for both roles.

4.3 Why the order matters

A qualified cover or the two-corner condition can take precedence over an immediate win elsewhere. This is deliberate: a non-losing strategy need not take every opportunity to win, and a winning switch still costs an active move. In the even-height uncertified phase, priority covering prevents a second active switch.

The fallback score counts permanently blocked bottom lines, ignoring out-of-board diagonals. It does not estimate future win probabilities. The reference checker enumerates current lines and targets; it is not a recursive game solver. With constant-time cell access, a straightforward checker considers $O(H)$ lines and $O(H)$ targets and can make a decision in $O(H^2)$ time. This is an estimate under that access model, not a proved machine-cost bound for the supplied Lean reference implementation.

5 First-player correctness for all finite heights

5.1 After certification

A first-player certificate is A. Lemma 2 excludes the next opponent first win. A legal cover is qualified and Step 1 chooses it. After a column fills, a qualified reply exists and Step 2 chooses one. Its own certificate, or safe terminal result, suffices regardless of which witness was used to prove existence. Once certified, the first player never switches actively again.

5.2 Small and odd heights

Height zero terminates immediately. At heights one, two, and three, the opening blocks the bottom horizontal; there are no diagonal or vertical fours and no opponent exceptions, giving A0.

For odd $H \geq 5$, the opening blocks the bottom horizontal and rising bottom diagonal. The only unresolved line is

$$D = ((1, 4), (2, 3), (3, 2), (4, 1)).$$

The target $(3, 2)$ has gap two and even outside capacity $3H - 1$, giving A1. Thus all these first-player cases start certified and use no active switch.

5.3 The first uncertified even-height family

For even $H \geq 4$, an uncertified first-player state may have the form

$$[F(SF)^a, \varnothing, \varnothing, (SF)^d], \qquad a, d \geq 0, \quad a \geq 1 \Rightarrow d \geq 1.$$

The root has $a = d = 0$. Only the first bottom corner belongs to F. Every opponent move leaves an empty middle column, excluding nonvertical wins, and exterior opponent stones are separated.

If S enters column two, adding F in column one is a qualified witness. The resulting potential exception $(2, 2)$ has its horizontal blocked by $(1, 2) = F$ if $a = 0$, or by $(4, 2) = F$ when $a \geq 1$ and hence $d \geq 1$. Its other relevant line and the bottom rising line are blocked by $(1, 1) = F$. If $a = 1$, the new $(1, 4)$ blocks D and gives A0; otherwise $(2, 3)$ is an A1 target of gap two with even outside capacity. The reply is legal because the old first-column height is odd and less than even H.

If S enters column three, covering at $(3, 2)$ blocks D and gives A0. If S fills column one, replying at $(2, 1)$ gives A1 with target $(2, 3)$; no new exception comes from the supported full top stone. These are existence witnesses, not requirements that the selector pick that exact column.

For an ordinary non-filling outside move, a qualified cover or other qualified reply may enter the certified regime early. If none exists, Step 3 has the following exhaustive cases:

- At opponent row at least three, covering preserves the family.
- A first move at $(4, 1)$ occurs here with $a = d = 0$. Every candidate still blocks the same two bottom lines, so the cover wins the tie, giving $d = 1$.
- A move at $(1, 2)$ with $d \geq 1$ admits no additional bottom-diagonal blocker in one legal reply; scores tie and the cover gives $a = 1$.
- The remaining low-row event is $S(1, 2)$ at the root. Only $F(4, 1)$ blocks all three bottom lines, so Step 3 chooses it and enters the second family.

At height four, Step 2 may already find a qualified reply and need not follow the last transition. Thus remaining uncertified either uses covers or spends one active switch to reach the two-corner family. Any other active reply directly enters certification.

5.4 The two-corner uncertified family

The second family is

$$[FS(SF)^a, (SF)^b, (SF)^e, F(SF)^d].$$

The first three heights are even and the fourth odd, and both bottom corners belong to F. Only column four can be filled by S. The bottom lines are blocked. The sole actual or potential exception is $(1,3)$: its horizontal is protected by $(4,3)$, empty or F, and its other valid line

$$U_1 = ((1,3),(2,4),(3,5),(4,6))$$

is protected by $(2,4)$, empty or F. Vertical runs stay short, and support descent excludes other first wins.

When the opponent's column remains nonfull, the two-corner clause forces a cover, even if another column offers a win. It preserves the family or enters A; no second active switch is spent. If neither termination nor certification occurs earlier, finite capacity forces a fill in column four. Stop if the board is full. Otherwise the following table supplies a qualified reply:

Conditions	Witness
$H = 4$	Any legal reply gives A0; U_1 is invalid.
$H \geq 6, a = 0$	$(1,3)$ eliminates the potential exception, giving A0.
$H \geq 6, a \geq 1, b \geq 2$	Any legal reply gives A0; $(2,4) = F$.
$H \geq 6, a \geq 1, b \leq 1, e = 2$	$(3,5)$ blocks U_1, giving A0.
$H \geq 6, a \geq 1, b \leq 1, e \leq 1$	Add in column three; $(3,5)$ is an A1 target.
$H \geq 6, a \geq 1, b = 1, e \geq 3$	$(2,3)$ wins immediately on D.
$H \geq 6, a \geq 1, b = 0, e \geq 3$	$(2,1)$ gives A2 with winning target $(2,3)$.

These disjoint cases exhaust the parameters. When $a \geq 1$, the actual exception's horizontal is blocked by the filled column's $(4,3) = F$. In the $e \leq 1$ case the new gap to $(3,5)$ is $4 - 2e$, positive and even, and the outside capacities are even. In the $e \geq 3$ cases, $(1,4), (3,2), (4,1)$ are fixed F stones, while the square $(2,4)$ required by U_1 lies above the winning target. All specified replies lie at row at most five and fit in $H \geq 6$.

The table establishes that Step 2 has a qualified option; the actual selector still uses its common priority rule. All these replies follow a filled column and are passive.

5.5 Induction

Small and odd heights start certified. At even heights, each uncertified reachable state belongs to one of the two families. Every opponent move is safe before the reply; the selector then preserves a safe family or enters certification, and a full-column event always has a qualified response. Finite capacity precludes endless waiting. The first-player rule is non-losing.

The first family can spend an active switch only to enter certification or the second family. The second family and the certified regime spend no more. The active budget is at most one, and is zero at odd heights and heights at most three.

6 Second-player correctness for all finite heights

6.1 Initial qualified replies exist

The following cases prove existence for the shared checker; they are not runtime opening tables.

At $H = 1$, the opening column is full. Any other bottom reply blocks the only horizontal and gives A0; the two remaining empty columns also give B.

At odd $H \geq 3$, an exterior opening can be covered, giving C. Since a qualified cover has first priority, the algorithm chooses it.

For an odd-height opening in column two, $(1,1)$ is a B witness with $R = \{3,4\}$. The bottom horizontal and rising diagonal are blocked. The falling diagonal is protected by $(2,3)$, outside R. The potential exception $(2,2)$ has its horizontal protected by $(3,2)$ and $(4,2)$; whichever odd column receives a later transfer, the other guard survives. Reflection handles column three.

For positive even height, reply at the bottom of the opening column's partner in $\{1,2\},\{3,4\}$. The initial two columns have odd capacity and the other adjacent pair has even capacity. It suffices to check openings in columns one and two, with reflection covering the rest.

After $F1, S2$, the bottom horizontal is blocked. The two bottom diagonals can be guarded by $(4,4)$ and $(3,2)$. The potential exception $(1,2)$ has its horizontal guarded by $(3,2)$ and its rising line by $(3,4)$. These guards lie in empty even-capacity columns, outside either transfer column. After $F2, S1$, the bottom horizontal and rising diagonal are blocked; $(3,2)$ guards the falling diagonal and, with $(4,2)$, the exceptional horizontal. The other relevant exceptional diagonal is already blocked at $(1,1)$. At $H=2$ only horizontals require inspection. Thus B holds in every even-height case.

6.2 The selector remains certified

In A and B, non-filling moves have qualified covers, selected by Step 1, and filling moves have qualified replies, selected by Step 2. In C, ordinary outside moves have qualified covers, while a first middle entry or outside fill has a B witness by Lemma 4.

The exterior bottom stones of a C state are the opponent's, so the two-corner clause cannot force the second player to cover wrongly after a middle entry. In A and B a cover was already qualified. Before the initial second-player move there are no second-player corners. Therefore the second player is certified after its first reply and never invokes Step 3. Each opponent move is first checked safe under the old certificate; only then is closure used.

6.3 Why the total count remains at most two

After every second-player reply the total remaining capacity is even. On first entering A or B there are at most two odd-capacity columns. To see this, at a positive even-height opening, a cover leaves all capacities even but no second-player bottom stone; the unblocked bottom horizontal has no positive-even-gap bottom target, so neither A nor B accepts, and C does not apply. A qualified reply must therefore switch, leaving two odd capacities.

At an odd-height interior opening, a cover leaves four odd capacities, no second-player bottom stone, and a nonempty middle column; A, B, and C all fail. Thus a qualified opening again switches and leaves two odd capacities.

An odd-height exterior opening enters C by covering. Ordinary C covers retain four odd capacities and no second-player bottom stone. A0 fails; an A target has odd outside capacity, so A1/A2 fail; B has the wrong number of odd columns. A middle-entry cover similarly leaves four odd capacities and no bottom block, and is not C; the other empty middle column also excludes an immediate winning cover. Qualified event replies must switch, turning two odd capacities even. An outside fill followed by an off-column reply also leaves two. At height one the first reply leaves two odd empty columns.

After entering A or B, ordinary covers preserve every capacity parity. If the opponent fills a column, the old odd set had two members, and there is a unique other odd-capacity column. B guarantees a qualified reply there. A0 allows it as well. For A1/A2, even total and even outside capacity imply that the target column has even capacity, so the other odd column is outside the target and gives a safe reply.

If no immediate winning reply exists, Step 2′s odd-capacity preference therefore chooses that unique odd column, leaving all capacities even. No later opponent move can fill a playable column before a cover. If a winning reply is selected instead, play stops and no further count is needed.

An active switch can occur at the opening or, after an exterior opening cover, at the first C middle-entry event; these do not occur on the same path. The opening or C exit contributes at most one total switch, followed by at most one passive switch in A/B. Thus the second player's active budget is at most one and its total count at most two. At $H=1$ the opening switch is passive. Choosing an arbitrary qualified post-fill move could retain safety while invalidating this particular count proof.

7 The uniform-response theorem

Theorem 5. For every finite $H \geq 0$, with immediate termination at $H=0$ and the first-player opening at $(1,1)$ otherwise, the shared response function of Section 4 is non-losing for each player against every legal opponent continuation. Each active budget is at most one. Total switches are at most four for the first player and two for the second.

Proof. Non-loss and active counts follow from Sections 5-6. Each passive reply follows a distinct newly full column; when all four are full there is no reply, so the first player has at most three passive switches. The stronger second-player bound was proved in Section 6.3. When both non-losing strategies are used, finite capacity forces a no-win full-board draw. ▫

Theorem 6. This algorithm attains the minimum worst-case active budget among all non-losing strategies:

Height	First player	Second player
$0, 1$	0	0
$2, 3$	0	1
Odd $H \geq 5$	0	1
Even $H \geq 4$	1	1

This is an optimality statement about active cost, not total switches, formula length, or the number of certificates.

8 Interface to the lower-bound proof

The companion mathematical paper [4, §7] proves that a second-player zero-active strategy loses at every $H \geq 2$, by successive first-player bottom moves in all four columns. It also proves the first-player lower bound at every even $H \geq 4$, with separate exterior-opening and interior-opening traps. Those traps allow all history-dependent passive replies after fills; they do not merely defeat a chosen greedy rule. In the interior trap, preparation of column one to exactly height four is essential.

These complete parametric lower bounds, together with our upper bounds, prove Theorem 6. They remain in the mathematical paper rather than being duplicated here. The Lean history-budget equivalence makes the strategy-class scope explicit, including arbitrary complete-history policies.

9 Conditions that cannot simply be deleted

9.1 A counterexample to omitting ordered B exits

At $H = 3$, consider the first-player turn

$$[FF, SFS, SFS, \varnothing].$$

It is reached legally by $F1, S2, F1, S3, F2, S2, F3, S3$, without an earlier win. For the second-player certificate, the odd-capacity columns are one and four, and the only unresolved line is row two. The empty $(4, 2)$ has gap two and is in the old G. A checker omitting B's ordered-exit condition would accept.

But F now fills column one, S transfers to $(4, 1)$, and $F(4, 2)$ completes the second-row horizontal. The old guard was lost when the transfer entered column four. The correct condition asks for protection outside column four or at its top $(4, 3)$ and rejects this position.

9.2 Safe choices need not produce the same trace

For the legal height-three prefix $F1, S2, F2, S2$, column two has just filled. A selector taking the leftmost qualified reply picks $(1, 2)$; the rule of this paper prefers an odd-capacity column and picks $(3, 1)$. Both are permanently safe, but they are different moves.

Safety alone therefore does not establish trace equivalence between selectors. We prove this rule's safety and budget, without claiming equality with every allowed choice in the companion mathematical construction.

10 Alternating segments and a linear target scan

10.1 At most nine segments in a candidate board

A maximal alternating segment is a maximal interval of a column whose adjacent stones have different colors. Equal-color neighbors create a boundary. Thus total segment count K equals the number of nonempty columns plus the number of equal-color adjacency pairs; segments are not monochromatic runs.

Proposition 7. Consider legal alternating gravity play from an empty width-w board, $w > 0$. A controlled player has so far made $A \leq a$ actual active and B actual passive switches. Immediately after a nonterminal opponent move, every legal candidate reply produces at most

$$K' \leq 2w - 1 + 2a$$

maximal alternating segments.

Proof. The controlled player's equal-color pairs number at most $A + B$: covering creates none, and a free opening creates none. Charge each opponent equal-color pair to the controlled reply immediately after its lower stone was played and left uncovered. That column was nonfull, since a later upper stone exists, so the charged reply was active. Different pairs have different lower stones and hence different charged replies. This includes any pair formed by the opponent's latest move. Therefore, before the candidate,

$$K \leq w + 2A + B.$$

Each passive switch corresponds to a distinct full column, and a reply is still required, so $B \leq w - 1$. If $B \leq w - 2$, the candidate adds at most one segment and

$$K' \le w + 2a + (w-2) + 1.$$

If $B = w - 1$, at least $w - 1$ columns are already full. The sole playable column is the opponent's last column, so every legal candidate covers an opposite-color stone and adds no segment. The same bound follows. We never impose the actual policy's budget on an unselected hypothetical candidate. ▫

For $w = 4, a = 1$, this gives nine. The bound is attained at height six along the legal sequence of column numbers `1,2,1,1,1,2,2,2,2,2,1,3,3,3,3,4,4,1,3,3,4`

With P first, its final board is

$$[PPQPPQ, QQPQPQ, QPQPPQ, QPP].$$

It has nine segments, one active switch, and three passive switches, and no prefix has ended the game. The supplied coordinate checker verifies this witness. It is an actual response state and was also a legal candidate. It prevents a uniform improvement to eight for this rule, but does not prove that all non-losing strategies need four total switches.

A bound obtained by separately adding a possible opponent segment and a candidate segment to a post-response estimate is loose. Counting at the opponent-move boundary and using the number of full columns gives the sharper bound above.

10.2 A2 targets as interval intersections

Fix a target column j of height h_j. A target row g satisfies

$$h_j + 2 \le g \le u, \qquad g \equiv h_j \pmod 2,$$

where u is bounded by H and by the lowest intersection of unresolved lines with column j. If an unresolved line has no such intersection, this target column cannot protect it. A target at gap at least two cannot complete a vertical line with three already fixed stones, so only slopes $\delta \in \{-1, 0, 1\}$ need checking.

An alternating segment $[l, r]$ in another column i contains P stones at one row parity. A target line needs P at $g + \delta(i - j)$, so that segment yields the interval

$$[l - \delta(i-j), r - \delta(i-j)]$$

for g, with a specified parity. Filter segments incompatible with $g \equiv h_j \pmod 2$. Each of the other three columns now supplies an ordered disjoint interval stream. Intersect the streams and the target range using three pointers.

At the current intervals let L be the maximum lower endpoint and R the minimum upper endpoint, including the target range. Round L upward to the required parity. If the rounded value is at most R, return that target. Otherwise advance an interval with minimum upper endpoint, or all tied such intervals. No discarded interval can participate in an omitted solution with a later, higher interval of another stream. Every pointer moves only forward, giving $O(K)$ work for fixed j, δ; four columns and three slopes add constant factors.

This replaces cubic enumeration of triples of segments for the A2 target search. It does not by itself make the complete checker linear: unresolved-line construction, deduplication, and other certificate queries remain separate tasks.

10.3 Input model and implementation evidence

The interval-scan subroutine agreed with independent cellwise enumeration in 80,020 comparisons, covering exhaustive small column-word combinations and random taller words. A direct compressed input of height 10^{12} with only three segments was also checked. These tests validate the implementation over those inputs; they are not an all-input refinement proof.

Along an accepted strategy, a constant segment bound permits storage in a constant number of height-sized words. Reading a raw H-row board is a different task and cannot ignore its input length. A height-sized word needs $\Theta(\log H)$ bits, so a constant count of word operations is not a claim of constant bit complexity.

The delivered executable optimization is the A2 interval-scan subroutine. A complete integrated compressed engine, equivalence of all its decisions to the accepted reference rule, and a machine-cost theorem remain unfinished.

11 Formal scope and reproduction

11.1 Model, histories, and costs

The project uses a standalone finite four-column model with relative colors P, Q. The recursive budget semantics checks every legal opponent move, first-win and full-board stopping, and the actual chosen reply, including the charge on a winning reply.

The module *HistoryBudget.lean* defines reachable legal prefixes and proves *budgetSafe_iff_historyBudget*. The *historyController* construction can carry the entire column history, without a finite-memory or computability assumption on its policy function. Thus the exact lower bounds are not confined to the stateless implementation.

11.2 Accepted roots and unfinished components

Result	Main source	Acceptance scope
Shared response, both roles, every finite height	*UniformFour/ FirstRoot.lean*, *SecondRoot.lean*	Parametric safety accepted
Exact active budgets	*FirstExact.lean*, *SecondExact.lean*	Upper and lower bounds accepted
History semantics and exact history budgets	*HistoryBudget.lean*, *HistoryRoots.lean*	Equivalence and roots accepted
Release parity, two-odd transfer, interval parity	*ReserveArithmetic.lean*	Arithmetic lemmas accepted
Nine-segment bound	*SegmentBoundArithmetic.lean*	Final arithmetic only; event injection and full-column premises remain explicit
General multi-target lemma and total-switch bounds	Companion mathematical paper	Paper proofs
Draws below four columns and at infinite height	*proof/Narrow*	Separate verified model; not part of this finite algorithm
Equivalence between the Covering and Narrow board models	Between the two projects	Not formalized
Complete compressed engine and its cost	Section 10	No full refinement or complexity theorem

The exact history roots are *UniformFour.U1.first_exact_history_budget* and *UniformFour.U1.second_exact_history_budget*. The shared executable response is *UniformFour.U2.respond*, defined through the finite checker. Selection order must be preserved to reuse the proven cost theorem.

11.3 Validation

The accepted toolchain is Lean 4.34.0-rc2; [3] describes Lean 4, while exact version and executable hash are fixed by the receipts. The source uses Lean's bundled libraries, with no Mathlib or large search certificate. All 55 modules compiled. Fourteen selected axiom closures contain only *propext*, *Classical.choice*, and *Quot.sound*. A trust-zero replay of the complete local import closure of *Audit* exited successfully. Every source and compiled artifact matched its recorded SHA-256 before and after replay.

Auxiliary checks include 396 coordinate instances of the first-player lower bound, 199 of the second-player lower bound, and a rerun of the supplied reference-rule checker for heights zero through twelve in both roles, totaling 83,429 states and 238,014 opponent edges. That rerun reused supplied code; only the added instrumentation and separately identified new checkers are new implementations. Seventy boundary cases compare the actual Lean response with the reference rule. None of these finite tests replaces the parameterized theorem.

From the repository root, run:

```
python tools/verify_proof.py
python tools/verify_proof.py --build --replay
```

The first command checks published source hashes and acceptance receipts; it does not run Lean. The second builds and kernel-replays the 55-module *proof/Covering* project. Install the version in its *lean-toolchain* and set *LEAN_BIN* if needed. Each Lean process uses all detected logical CPUs.

The separate *proof/Narrow* project supplies finite-width-one-to-three and infinite-height-width-one-to-four draw theorems. Its four mathematical modules and audit entry were freshly compiled, six selected axiom closures satisfied the same whitelist, and a trust-zero import replay succeeded. It needs the pinned Mathlib dependencies. Once these are installed, run *python tools/verify_proof.py –supporting –build –replay*. No equivalence between the two board models has been formalized. Full instructions and the source manifest are in *docs/verification.md* and *audit/*.

12 Implementation boundaries and next steps

The deliverables are the accepted finite-board reference rule, the segment-count argument, and the A2 scan subroutine. A complete compressed implementation is not presented as finished software. The next refinement should first establish agreement of compressed and cellwise board access, drops, lines, and certificates; then prove that the selector returns the same column; finally account for the chosen machine model.

Order changes require their own checks. Immediate-win priority, arbitrary qualified full-column transfers, or removal of an ordered-exit condition may invalidate the established cost or safety proof. Implementation work in this paper does not change the companion paper's status as an independent mathematical strategy proof.

References

[1] L. Victor Allis. *A Knowledge-based Approach of Connect-Four: The Game is Solved: White Wins*. Master's thesis, Vrije Universiteit Amsterdam, Report IR-163, 1988. Especially §§2.2, 6.1, 7.1. Full text.

[2] Yoshiaki Yamaguchi, Kazunori Yamaguchi, Tetsuro Tanaka, Tomoyuki Kaneko. Infinite Connect-Four Is Solved: Draw. In *Advances in Computer Games*, LNCS 7168, 2012, pp. 208-219. doi:10.1007/978-3-642-31866-5_18.

[3] Leonardo de Moura, Sebastian Ullrich. The Lean 4 Theorem Prover and Programming Language. In *Automated Deduction - CADE 28*, LNCS 12699, 2021, pp. 625-635. doi:10.1007/978-3-030-79876-5_37.

[4] GPT. *Draw Strategies for Gravity Connect Four on One to Four Columns: Covering Strategies, Optimal Active-Switch Budgets, and Lean Verification*. Companion mathematical paper, 8 November 2026. PDF.

[5] GPT. *UniformFour: Lean definitions and proofs*. Source. 2026.

[6] GPT. *Interval scanning and coordinate checks*. Source. 2026.